

IBI-DT: a novel approach combining individualized Bayesian inference and decision tree for identifying cancer drivers and their interactions

Md Asad Rahman^{1,2}, Gregory F. Cooper³, Jinying Zhao², Xinghua Lu³, Jinling Liu^{2,*}

¹Department of Engineering Management and Systems Engineering, Missouri University of Science and Technology, 600 W 14th St, Rolla, MO 65409, United States

²Department of Epidemiology, College of Public Health and Health Professions and College of Medicine, University of Florida, 2004 Mowry Road, Gainesville, FL 32610, United States

³Department of Biomedical Informatics, University of Pittsburgh, 5607 Baum Blvd, Pittsburgh, PA 15206, United States

*Corresponding author. Department of Epidemiology, College of Public Health and Health Professions and College of Medicine, University of Florida, Gainesville, FL, USA. E-mail: jinling.liu@ufl.edu

Abstract

Cancer is mainly caused by a relatively small portion of somatic genome alterations (SGAs), called cancer drivers. Despite success in identifying a good number of cancer drivers, many more remain to be discovered to explain various cancers. Moreover, limited tools are available to identify potential interactions among cancer drivers for a better understanding of oncogenesis. To tackle these challenges, we have developed a novel approach called individualized Bayesian inference using a decision tree (IBI-DT). IBI-DT recognizes the genetic heterogeneity among cancer patients, where different individuals or patient subgroups of distinct genomic makeup may have different drivers. IBI-DT works by constructing smaller subgroups with similar genetic makeup (i.e. patient-like-me subgroups) using a decision tree structure and analyzing multiple trees to identify the SGAs that play a significant role in regulating downstream gene expression patterns at the subgroup and individual levels. This is distinct from population-based approaches, which tend to evaluate the influence of an SGA for the entire population, thereby likely missing low-frequency SGAs that may well explain a small subgroup of cancer patients. Also importantly, IBI-DT can efficiently identify cancer drivers that may have functional interactions. We applied IBI-DT to identify cancer drivers regulating the downstream differential gene expression in cancer patients and compared it to the standard, population-based method of expression quantitative trait loci analysis. Our results show that IBI-DT performs well in identifying both important cancer drivers, especially the low-frequency drivers, and their interactions, allowing for a better understanding of the cancer signaling pathways.

Keywords: individualized Bayesian inference; decision tree; somatic genome alterations; cancer driver; genetic interactions

Introduction

Human cancers are primarily driven by somatic genome alterations (SGAs), such as mutations and copy number changes [1]. While hundreds of SGAs can be present in a tumor, most are passengers, with only a subset acting as true drivers of tumorigenesis [2]. Identifying these cancer drivers is critical for understanding cancer mechanisms and guiding therapy. As of September 2023, the Cancer Gene Census (CGC) lists 738 known drivers [3], but many remain undiscovered, particularly in rare cancers and genetically distinct patient subgroups, highlighting the need for new, more sensitive discovery methods.

Current cancer driver identification methods fall into several categories. The most common are frequency-based models such as MutSigCV, which identify SGAs due to their elevated mutation rates across cohorts [4, 5]. However, these approaches rely on estimates of background mutation rates and, more importantly, do not directly assess the functional impact of SGAs, such as their regulatory effects on gene expression or influence on signaling

pathways. To address these limitations, a second class of methods incorporates gene expression and network context to infer the downstream consequences of SGAs. DriverNet links SGAs to differentially expressed genes (DEGs) through an influence graph [6], while HotNet2 identifies mutated gene modules using heat diffusion in protein interaction networks, with gene expression used post hoc for interpretation [7]. PhenoDriver introduces some personalization by associating SGAs with pathway-level expression changes [8]. More recently, deep learning methods like HGDCs and CGMega integrate multi-omics using graph neural networks, typically treating expression as static input features [9, 10]. Although these methods utilize expression data and sometimes attempt personalization, they generally do not model direct SGA → DEG regulation and often depend on prior networks that may be incomplete or biased. Most importantly, they focus on cohort-level prioritization rather than subgroup- or individual-level analysis, which limits their ability to capture genetic heterogeneity across tumors and

Received: March 4, 2025. Revised: July 7, 2025. Accepted: August 11, 2025

© The Author(s) 2025. Published by Oxford University Press.

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (<https://creativecommons.org/licenses/by-nc/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

detect low-frequency drivers relevant to specific subgroups or individuals.

Recognizing the importance of modeling functional regulation and heterogeneity, Cai et al. developed the tumor-specific causal inference (TCI) method to directly assess the likelihood that an SGA regulates a DEG in a tumor or subgroup [11]. It estimates a causal score of an SGA based on its influence on DEGs in a tumor, enabling the identification of functional SGAs that impact the molecular systems of cancer cells. TCI marked a shift toward functionally grounded, individualized analysis of driver events. Building upon this concept, we introduced the individualized Bayesian inference (IBI) framework, which generalizes the TCI strategy into a flexible probabilistic model for scoring genomic variants [12]. Unlike TCI, which is limited to somatic mutations (SM) and gene expression in cancer, IBI is designed to accommodate a broad range of applications—including both molecular and clinical phenotypes, somatic or germline genomic data, and disease traits beyond cancer.

To better account for population-level genetic heterogeneity and uncover functional interactions among genomic variants, we developed IBI-DT, a generalized method of IBI using decision trees. Unlike IBI or TCI, which evaluate single SGA → DEG relationships or perform binary partitioning, IBI using a decision tree (IBI-DT) recursively divides the population into genetically coherent subgroups. At each node, SGAs are selected by maximizing the Bayesian marginal likelihood, which quantifies how well each SGA explains DEG variation in the “patient-like-me” subgroup. This tree structure enables principled ranking of both individual SGAs and their interactions. SGAs that co-occur along a branch are inferred to regulate the same DEG either in conjunction (when the upstream SGA is mutated) or disjunction (when the upstream SGA is unmutated). These expression-informed relationships distinguish IBI-DT from statistical co-mutation methods such as DISCOVER or MEMO [13, 14], which rely solely on mutation pattern enrichment and do not model downstream phenotypic effects. IBI-DT also differs fundamentally from classical decision tree algorithms and random forests, which focus on phenotype prediction using heuristic splitting rules, such as Gini impurity or information gain [15–20]. In contrast, IBI-DT is explicitly designed for discovery, employing marginal likelihood as a Bayesian decision rule to identify SGAs with functional regulatory effects, thereby offering interpretable and biologically grounded inferences.

To evaluate IBI-DT's ability to detect SGAs of potential functional impact, we benchmarked it against expression quantitative trait loci (eQTL) analysis [21, 22], a standard population-level method that models SGA → DEG associations using linear regression for continuous traits and Fisher's exact test for discrete traits [23]. We selected eQTL as a baseline because it shares the same input data (SGA and DEG) and inference target (regulatory association), without relying on external network or pathway priors. This makes it a strong and appropriate control for IBI-DT. Our results demonstrate that IBI-DT outperforms eQTL in identifying both known and novel cancer drivers, especially low-frequency SGAs with subgroup-specific effects, and recovers biologically plausible interaction pairs missed by existing methods. By combining principled Bayesian inference, patient-like-me subgroup modeling, and decision-tree-based interpretability, IBI-DT provides a powerful framework for personalized, functional discovery of cancer drivers and regulatory mechanisms.

Material and methods

Dataset and selection of differentially expressed genes

We followed the TCI paper to prepare both SGA and DEG data [11]; please refer to the TCI paper for the detailed processing procedure. Briefly, the SGA data are a combination of SM and somatic copy number alterations (SCNA) that resulted in an altered gene in a tumor defined by either an SM or SCNA event, totaling 17,646 SGAs. SM data were obtained directly from the TCGA portal (<https://tcga-data.nci.nih.gov/tcga/dataAccessMatrix.htm>) that contained 16 distinct cancer types and 5097 total samples upon then; the sample size of each cancer type is included in [Supplementary Table S1](#). We considered all nonsynonymous SM across genes, with genes classified as mutated if they contained one or more nonsynonymous SM or indels. SCNA data were downloaded from the Firehose browser of the Broad Institute (<http://gdac.broadinstitute.org/>), for which we only considered homozygous deletion or high copy number amplification. For DEG data, we retrieved gene expression information from the Broad Institute. DEGs were identified by comparing the expression of tumor tissue to those from a group of adjacent “normal” tissues using Gaussian distribution and calculating p-values. DEG data is represented by a tumor-by-gene binary matrix, indicating whether a gene is differentially expressed in a tumor. The procedure included the removal of DEGs linked with SCNA events or highly correlated with tissue origin.

In this investigation, as a proof of concept, we used DEG MPL as the trait to investigate the performance of IBI-DT, relative to eQTL, in identifying cancer drivers that drive cancer formation via regulating the downstream gene expression patterns; we further replicated and validated the findings with another DEG JUN. MPL encodes the thrombopoietin receptor, and its dysregulation, primarily at the expression level, has been implicated in aberrant PI3K signaling in hematologic malignancies [24]. JUN is a core component of the AP-1 transcription factor complex and is frequently overexpressed across diverse solid tumors, where it drives pro-oncogenic transcriptional programs [25, 26]. We selected MPL and JUN as representative traits because they are well-established cancer-related genes with known upstream regulators. This prior knowledge allows us to use their known regulators as a gold standard for evaluating the inference accuracy of IBI-DT. Moreover, their regulatory landscapes are biologically interpretable and complementary, together spanning distinct cancer types and mechanisms. We also emphasize that there is no reason to believe the selection of MPL and JUN introduces bias into the evaluation. Importantly, IBI-DT is a generalizable method that can be applied to any binary trait, including genome-wide analyses of DEGs. Our goal in this study is to introduce and evaluate IBI-DT using well-characterized DEGs as proof-of-concept examples, with genome-wide or pathway-level applications planned for future work.

In this study, we use pan-cancer analysis to identify SGAs and their interactions that regulate specific DEGs, because (i) certain cancer drivers or interactions are shared by different cancer types, and we would like to discover the common impact of a cancer driver or its interaction across cancer types; (ii) pan-cancer analysis increases statistical power while a much smaller sample size is available for a particular cancer type. Future applications of IBI-DT to individual tumor types will be important for identifying cancer-type-specific drivers and their interactions as larger cohorts become available.

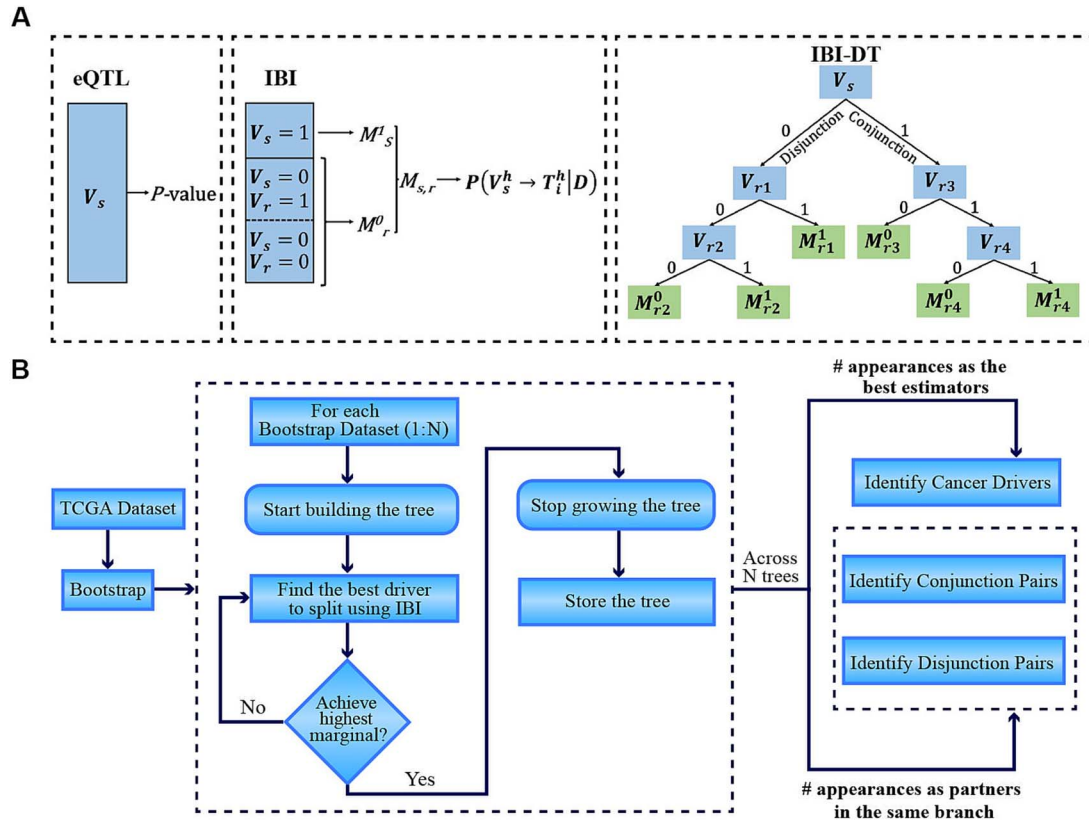


Figure 1. The IBI-DT method and pipeline. (A) Schematic of the methods of eQTL, IBI [12] and IBI-DT. M represents marginal likelihood. The IBI schematic is adapted from Rahman et al., BMC Genomics (2023), of which the authors retain copyright, published under CC BY 4.0. (B) The IBI-DT pipeline.

Expression quantitative trait loci

eQTL analysis is commonly employed to model the relationship between genetic variations and gene expression levels. As a population-based approach, eQTL analysis considers all samples within a population to evaluate the influence or effect of a genomic variant (Fig. 1A). Depending on the nature of the data, various statistical methods may be employed in analysis. For continuous gene expression levels, linear regression is typically used, whereas Fisher's exact test can be applied for discrete data [21, 22]. Given the discrete nature of both SGAs and DEG data, we employed Fisher's exact test to model the relationship between them.

The IBI-DT framework

IBI-DT builds on the IBI framework [12], which models probabilistic relationships between genomic variants (e.g. SGAs) and phenotypic traits (e.g. DEGs) using a bipartite Bayesian network. In this structure, the nodes are divided into two disjoint sets, SGAs and DEGs, and edges are directed from SGAs to DEGs to represent candidate regulatory influences. To evaluate these edges, IBI uses the Bayesian Dirichlet equivalent uniform scoring function to compute the marginal likelihood, which quantifies how well a given edge or model explains the observed data by integrating over all possible conditional probability values under a Dirichlet prior [11, 12, 27]. IBI partitions the population into two subgroups based on the mutation status of a candidate variant V_s , and evaluates its influence on a target phenotype T_i by computing the marginal likelihood separately within the "patient-like-me" subgroup (sharing the same status of $V_s = 1$) and the complementary subgroup. These subgroup-level marginal likelihoods are then used to derive

the individualized posterior probability that V_s influences T_i for each individual (Fig. 1B).

IBI-DT generalizes this approach by recursively partitioning the population into finer subgroups through a decision tree structure. At each node, the algorithm selects the variant V_s that maximizes the marginal likelihood of explaining expression variation in the subgroup. The population is then split into two branches based on the mutation status of V_s , and the process repeats until further splits no longer improve the marginal likelihood. To avoid overfitting, IBI-DT uses a structure prior (Equations 1 and 2) to guide decision tree construction by penalizing overly complex trees with long paths. Equation 1 computes the Bayesian posterior probability of a model given data (D) from the marginal likelihood ($P(D | \text{Model})$), while Equation 2 defines the structure prior, where $|\text{Path}|$ is the path length and q between 0 and 1 controls the degree of penalty. Since $P(D)$ is constant, the log posterior probability $P(\text{Model} | D)$ is proportional to the score S in Equation 3. S combines the log marginal likelihood and the log structure prior penalty, and is used to evaluate nodes and decide whether further splits should be made during tree growth.

$$P(\text{Model} | D) = \frac{P(D | \text{Model}) P(\text{Model})}{P(D)} \quad (1)$$

$$P(\text{Model}) = q^{|\text{path}|} \quad (2)$$

$$S = \log[P(D | \text{Model}) \cdot P(\text{Model})] = \log[P(D | \text{Model})] + \log[P(\text{Model})] \quad (3)$$

To ensure robust inference, we employed a stratified bootstrap procedure [28] within IBI-DT. When applied to the TCGA data, we generated N ($N = 200$) bootstrapped datasets by sampling TCGA

tumors with replacement while preserving the class proportions of the target trait (e.g. DEG MPL or JUN). For each resampled dataset, an IBI-DT tree was constructed by recursively selecting SGAs that maximize marginal likelihood and recording all decision nodes, branch structures, and selected variables.

To quantify the importance of SGAs in regulating a downstream trait (e.g. a DEG), we computed two frequency-based statistics across the ensemble of 200 bootstrapped IBI-DT trees. For each SGA, we recorded the number of trees in which it appeared as a decision node which we denote as α . This count reflects how often an SGA was selected as the best explainer of gene expression variation within a subgroup, guided by the Bayesian marginal likelihood. Conceptually, α serves as a computationally efficient approximation to Bayesian model averaging (BMA) [29]: rather than summing over all possible trees, we bootstrap datasets, fit models under a Bayesian criterion, and tally each SGA's recurrence. While we explored theoretically grounded alternatives for BMA, such as importance sampling [30] for posterior model weighting, these were either less robust or substantially more computationally intensive. Empirically, the α statistic consistently outperformed other approximations in recovering literature-supported cancer drivers.

To detect regulatory interactions, we analyzed co-occurrences of SGAs along tree branches. SGAs appearing on the same tree branch or path are treated as candidate interaction pairs. If the parent node's SGA is mutated, it forms a conjunction with downstream SGAs; if unmutated, it forms a disjunction, suggesting alternate regulation. For each pair, we computed β , the number of trees in which both SGAs co-occurred in the same path, regardless of adjacency. Pairs with higher β are ranked as stronger candidates for functional interaction. Together, α and β support robust identification of both individual drivers and expression-informed interaction patterns.

These α and β statistics were also extended to the individual level to support personalized inference. For a given individual h , the individualized posterior probability for V_s influencing T_i , $P(V_s^h \rightarrow T_i^h | D)$, is calculated as the normalized α among SGAs mutated in that individual by taking into consideration the distinct genetic makeup of that individual (i.e. by dividing the α of the mutated SGA by the sum of α for all mutated SGAs present in this individual; Equation 4). It is derived using Bayesian model posterior formula (Equation 1), while assuming that only one variant V_s is influencing a given trait T_i at a time and the prior probability of each variant influencing a given trait is uniform.

$$P(V_s^h \rightarrow T_i^h | D) = \frac{\alpha(s)}{\sum_{k \text{ mutated SGAs in } h} \alpha(k)} \quad (4)$$

An analogous individualized probability can be computed for each SGA pair based on the β statistic, conditional on the SGA pairs co-occurring in the individual's mutation profile. These individualized probabilities provide interpretable, patient-specific (PS) insights into candidate regulatory mechanisms.

The pseudocode below outlines the process of building trees through IBI-DT, using as an example the identification of cancer drivers that regulate downstream gene expression. The computation efficiency of IBI-DT is highly optimized using GPU and matrix computation: it takes about 26 min on a NVIDIA Tesla V100 Tensor Core to derive all 200 trees of high-order interactions using IBI-DT from the TCGA dataset.

A. Input: a TCGA dataset of 5097 samples x 17,646 SGAs, and one binary trait T_i (e.g. DEG MPL).

B. Output: An ensemble of N trees (e.g. $N=200$), along with α and β statistics from the trees and individualized posterior probabilities.

C. Procedure for building bootstrapped trees:

1. Bootstrapping: Take a bootstrap sample with replacement, stratified by trait status.

2. Tree construction:

- I) Identify a variant V_s (e.g. an SGA) that maximize the score of S , relative to the other variants, i.e. best explaining a trait T_i for the current population.
- II) Split the population based on the mutation status of V_s (0 or 1).
- III) Recursively repeat steps I-II on subgroups until no split will improve the score.
- IV) Store all decision nodes and split paths.

3. Repeat steps 1–2 N times to construct an ensemble of N bootstrapped decision trees.

D. Driver inference (α). Count how often each SGA appears in any decision node across all trees. This frequency, α , reflects the variant's overall regulatory influence.

E. Interaction inference (β). For each SGA pair appearing on the same branch (regardless of adjacency), increment its β count. Interaction type is assigned based on parent mutation status (conjunction vs. disjunction).

F. Individualized inference. For a given individual h , calculate his or her individualized posterior probabilities for a single driver or an interacting pair based on his or her distinct genetic makeup and the calculated α/β statistics.

Random node experiment

To understand the significance of the α and β statistic, we conducted a random node experiment by using the tree structures derived from IBI-DT. In this experiment, original SGAs on all the nodes within the IBI-DT-built trees were replaced with randomly selected genes while the tree structure stayed unchanged. Throughout this process of replacement, an SGA was selected at random from the comprehensive list of 17,646 SGAs to replace the informative SGA selected by IBI-DT. This process resulted in 200 trees that maintained the structure integrity of the original IBI-DT trees but had randomly chosen nodes. Subsequently, we computed the α statistic for each SGA across all 200 trees, as well as the β , imitating the approach utilized in IBI-DT. The values of these α and β statistics from this random node experiment function as a null-hypothesis distribution for the IBI-DT experiment.

Patient-specific analysis

We performed PS analysis by calculating the individualized posterior probability based on each individual's specific genomic makeup (Equation 4). For a particular SGA with a unique α at the population level, it may have different individual-level posterior probabilities among different individuals, reflecting each individual's distinct genomic makeup. For a given individual having MPL differential expression (i.e. MPL = 1), IBI-DT ranks all the mutated SGAs present in this individual based on their individualized posterior probabilities; the SGA with the highest posterior probability is considered to have the most significant impact on the trait of interest for this particular individual. We referred to that SGA as the PS driver by IBI-DT. Additionally, we identified a top SGA for each individual with MPL differential expression based on the population-level p-values obtained through eQTL: among the mutated SGAs present in a given individual, the one with the

lowest (most significant) p-value was considered to most likely influence the *MPL* status in this specific individual. We refer to it as PS-driver by eQTL.

Results

In this study, we applied IBI-DT to estimate cancer drivers among 17,646 SGAs that regulate downstream DEGs in a TCGA dataset of 5097 tumor samples across 16 cancer types. By leveraging pan-cancer data, we sought to identify shared cancer drivers. As a proof of concept, we demonstrated IBI-DT's performance by examining drivers of the DEG *MPL* and validating them against another DEG, *JUN*. We compared our results with eQTL analysis, a standard method for identifying genomic factors that influence downstream gene expression. We selected *MPL* and *JUN* because they are well-known cancer markers with established upstream regulators, serving as gold standards to evaluate IBI-DT performance. Our main objective was to introduce and assess IBI-DT's capability in pinpointing cancer drivers and their interactions using specific and well-characterized DEGs, rather than exhaustively discovering all potential cancer drivers with genome-wide or pathway-level applications that will be addressed in future work. Results for DEG *MPL* are provided in the main text, while validation using DEG *JUN* is detailed in the Supplementary Information.

IBI-DT outperforms eQTL in ranking key drivers regulating the downstream gene expression

The PI3K/AKT signaling pathway is pivotal in regulating *MPL* expression [24]. Members of this pathway, including *PIK3CA*, *PTEN*, *AKT1*, and *PIK3R1*, act as key regulators to influence *MPL* expression levels (Fig. 2A). These well-known key drivers for DEG *MPL* can serve as a gold-standard reference for evaluating the performance of IBI-DT and eQTL.

To evaluate the capability of identifying drivers by IBI-DT, relative to eQTL, across smaller to larger sample sizes, we created 10 subsets from the full dataset of 5097 tumor samples using a stratified random sampling method. Each subsequent dataset incorporates approximately 500 more samples than the dataset before it, ranging from 500 to 5097 total. To maintain uniform frequencies for DEG *MPL* across all 10 datasets, we used stratified random sampling based on DEG *MPL* status. The rankings of key drivers were determined using both IBI-DT and eQTL across the 10 datasets. In the IBI-DT method, SGAs were ranked based on the α measure, with those appearing most often as the best drivers receiving the top rankings. eQTL determines the SGA rankings by the P-values, with those having the lowest p-values receiving the highest or best ranks. In the context of these rankings, a lower numerical value such as 1, 2, or 3 indicates a higher or top-ranking SGA, signifying higher significance. Conversely, higher numerical values represent lower or bottom rankings, indicating less significance in a dataset.

For all 10 datasets, IBI-DT consistently outperformed eQTL by assigning higher (better) rankings to the aforementioned well-known drivers. For example, our analysis shows that IBI-DT assigns significantly better rankings to *PIK3CA* when compared to eQTL. In the smallest dataset of 500 samples (Fig. 2B), IBI-DT ranked *PIK3CA* at 200, while eQTL placed it at a much lower rank of 675. As the sample size increased, IBI-DT maintained its superior ranking performance, ultimately positioning *PIK3CA* at the top, whereas eQTL ranked it at position 20 using the full dataset of 5097 tumor samples. For *PIK3R1* (Fig. 2C), IBI-DT initially ranked it low in the smallest sets, where it didn't appear as a node across

all the trees. However, as sample size increased, IBI-DT detected and steadily assigned better rankings, peaking at sixth place in the full set. As a comparison, eQTL started with a rank of 13,060 for *PIK3R1* in the smallest set and improved its position to 649 in the full dataset, but still far away from matching IBI-DT's performance. Our results also indicate that IBI-DT consistently outperforms eQTL in ranking the *PTEN* gene across all 10 sets. IBI-DT assigns lower rankings to *PTEN* (Fig. 2D), starting with 574 in the smallest set and improving to the rank of 2 in the full dataset. In contrast, eQTL ranked *PTEN* significantly lower, ranking 3953th in the smallest set and gradually improves to 173 in the full dataset. Similar to the trend observed with the other key drivers just discussed, IBI-DT outperformed eQTL in *AKT1* rankings (Fig. 2E). IBI-DT ranks *AKT1* at 162 in the smallest set, which is significantly better than eQTL's rank of 931. In the full dataset, IBI-DT placed *AKT1* at fifth place, while eQTL's rank position was at 645th. These results provide support for the better performance of IBI-DT in accurately identifying key regulatory genes in the PI3K/AKT signaling pathway that influence *MPL* expression.

IBI-DT found more well-known cancer drivers than eQTL, especially those with low mutation frequency

Considering that DEG *MPL* is a common cancer marker regulated by the PI3K pathway, drivers regulating *MPL* expression identified by IBI-DT and eQTL could be drivers involved in oncogenesis in general, influencing *MPL* and other DEGs through pathway cross-talks. Therefore, we further investigated whether the top drivers identified by IBI-DT for *MPL* expression contain well-known and novel cancer drivers in general.

To establish a robust threshold for top driver selection, we conducted a random node experiment as a control, yielding a maximum α value of 7 across all the SGAs. In our IBI-DT experiment, 402 SGAs notably deviated from the expected distribution of random occurrences by exceeding seven. Within this group, we narrowed our focus to the top 50 SGAs ($\alpha \geq 22$) for the follow-up analysis, where each SGA greatly surpassed the maximum threshold from the random node experiment, supporting their statistical significance and suggesting potentially biological significance (Fig. 3A).

The Catalogue of Somatic Mutations in Cancer (COSMIC) CGC is the flagship database for validated cancer drivers [3]. We used the CGC database that catalogs 738 validated cancer drivers including 579 well-established cancer drivers in Tier 1 and 159 emerging cancer drivers in Tier 2 as of September 2023. We evaluated IBI-DT's performance against eQTL and the random node experiment across 10 datasets (Fig. 3B). In our analysis, we consider both tiers, encompassing well-established cancer drivers as well as those with emerging evidence of their role in oncogenesis. As expected, the number of CGC genes returned by random trees fluctuated below 7 in the 10 datasets, not showing any higher power with increased sample sizes. By contrast, eQTL showed a moderately increased capability in identifying significant CGC cancer drivers with larger sample sizes. The performance of IBI-DT in revealing significant CGC cancer drivers improved dramatically with increased sample sizes. Moreover, for each of the 10 datasets except the smallest one, IBI-DT outperformed eQTL, by a large margin, in cancer driver discovery, according to this analysis. For example, using the largest dataset, among the top 50 SGAs, IBI-DT found 17 known cancer drivers while eQTL identified only five known cancer drivers. Overall, IBI-DT showed much better performance, relative to eQTL, in discovering well-known cancer drivers

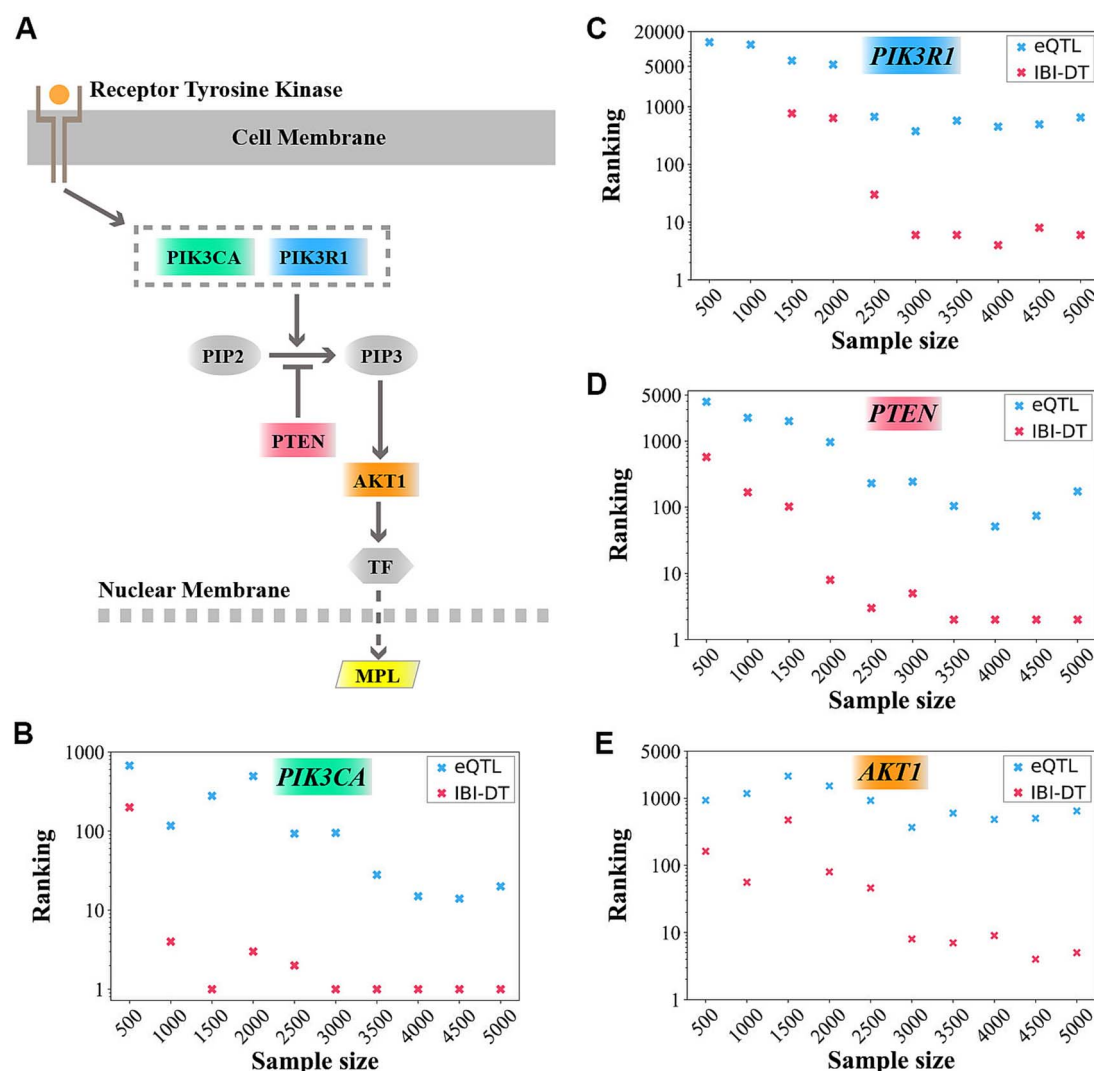


Figure 2. The ranking of well-known drivers for MPL expression by IBI-DT and eQTL. (A) PI3K/AKT signaling pathway regulates MPL expression. (B) *PIK3CA* ranking among 10 datasets of increasing sample size. (C) *PIK3R1* ranking among the 10 datasets of increasing sample size. (D) *PTEN* ranking among the 10 datasets of increasing sample size. (E) *AKT1* ranking among the 10 datasets of increasing sample size.

that potentially influence the expression of MPL, across 10 different datasets from smaller to larger sample sizes. To further evaluate IBI-DT's ability to identify cancer drivers, we expanded our evaluation to include two additional comprehensive databases—OncoKB (1185 genes) [31] and IntOGen (633 genes) [32]—alongside the COSMIC CGC. Among the top 50 SGAs identified by IBI-DT, 17 were known cancer drivers listed in CGC, 16 in OncoKB, and 15 in IntOGen. In contrast, the eQTL-based approach identified only five, seven, and four known drivers in the same databases, respectively. When combining all three sources, IBI-DT recovered 17 unique known cancer drivers, compared to seven identified by eQTL, supporting IBI-DT's better performance relative to three primary catalogs of cancer driver genes. For parsimony, in the main figures we focus on performance relative to the CGC. We examined the mutation frequency for the top 50 SGAs respectively ranked by IBI-DT and eQTL to evaluate their ability to identify low-mutation-frequency or rare drivers. Interestingly, among the top 50 SGAs, IBI-DT identified 17 SGAs with mutation frequencies at or below 0.01, whereas eQTL only identified three SGAs with such low mutation frequency (Fig. 3C). Notably, among IBI-DT's top 50 SGAs, 16 CGC drivers were missed by eQTL, including low-frequency drivers such as *AKT1* and *PIK3R1*, which IBI-DT

ranked as fifth and sixth, respectively, compared to 645th and 649th by eQTL (Fig. 3D). While IBI-DT is effective at detecting low-frequency cancer drivers, it identifies high-frequency CGC genes as well. For instance, *TP53* (MF = 0.44), *PIK3CA* (0.21), *PTEN* (0.11), *LRP1B* (0.15), *CDKN2A* (0.17), and *CSMD3* (0.18) were highly ranked by IBI-DT but missed or poorly ranked by eQTL. Conversely, eQTL identified four moderately frequent CGC genes (*BRIP1*, *PPM1D*, *IKZF3*, and *CLTC*) not ranked in IBI-DT's top 50. These results highlight that IBI-DT does not sacrifice high-frequency drivers and, by modeling regulatory effects within subgroups rather than mutation frequency alone, offers comprehensive discovery across a range of mutation frequencies.

Among the top 50 SGAs identified by IBI-DT, Table 1 shows that 17 are known drivers included in the CGC database. Some of the remaining 33 SGAs are potentially novel cancer drivers, but further research is needed to investigate which of them are true drivers.

IBI-DT identified potential drivers that better explain the MPL status in individual tumors

One goal of personalized oncology is to identify the specific driver(s) that drive cancer formation in a patient so that

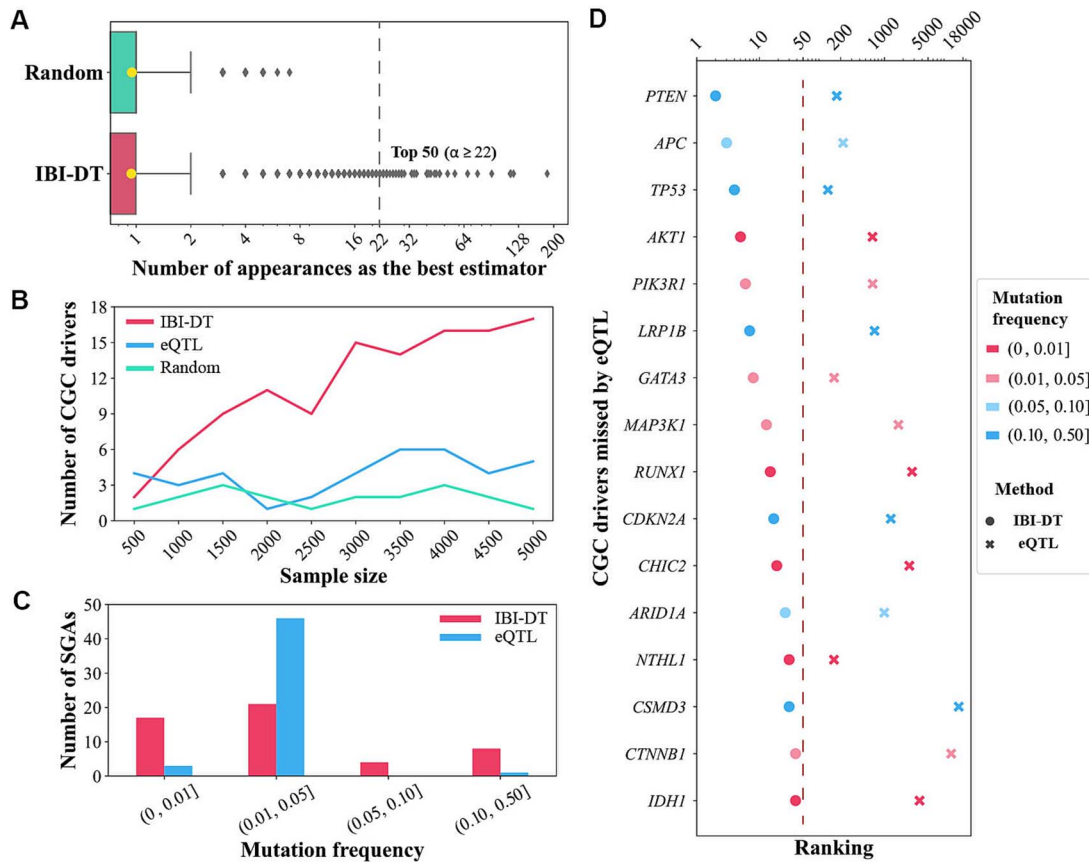


Figure 3. Top 50 significant SGAs identified by IBI-DT and eQTL respectively for DEG MPL as the target. (A) Boxplot comparison of the α for single drivers between the Random and IBI-DT methods. The vertical dashed line shows the α threshold of 22 for the top 50 SGAs. (B) Identifying CGC genes among 10 datasets using IBI-DT, eQTL, and Random. (C) Mutation frequencies of top 50 SGAs according to IBI-DT and eQTL. As an example, (0.0, 0.01) on the x-axis indicates the mutation frequency that is >0.0 but ≤ 0.01 . (D) CGC genes missed by eQTL but identified by IBI-DT when considering the top 50 SGAs identified by each method.

personalized treatment can be given to that patient. With that eventual goal in mind, we performed PS analysis to identify the top-1 SGA as the potential driver for each individual cancer patient with MPL differential expression ($MPL=1$), referred as PS-driver. IBI-DT identified 90 unique PS-drivers, while eQTL identified 201, to explain the 835 tumor samples with MPL differential expression (Fig. 4A). In general, the IBI-DT top SGAs ranked by α were able to better explain the MPL status in individual tumors, relative to the eQTL top SGAs ranked by P-value. In other words, with the same ranks by α or P-value, IBI-DT PS-drivers were able to explain many more MPL = 1 tumors, compared to eQTL PS-drivers. For example, the IBI-DT PS-drivers among the top-5 ranks explained 550 (66%) MPL = 1 tumors while the eQTL PS-drivers among the top-5 ranks only explained 70 (8%) MPL = 1 tumors (Fig. 4A). At rank 100, the patient coverage by IBI-DT PS-drivers increase to 796 (95%) while the patient coverage by eQTL PS-drivers is only 481 (58%) (Fig. 4A). With this PS analysis, PIK3CA was recognized by both IBI-DT and eQTL, as the PS-driver explaining the most tumors with MPL = 1: $n=248$ by IBI-DT and $n=182$ by eQTL. PIK3CA was ranked 20th in the eQTL experiment when using population P-values. This may be interpreted as the eQTL method better identified PIK3CA (first in terms of explaining the most tumors) using a more individualized approach, compared to its 20th rank using a population-level P-value. The top SGA identified for each individual tumor by IBI-DT shows an individual-level posterior probability ranging from 0.01 to 0.92 (Fig. 4B). In our data, a particular (MPL = 1) tumor

can have up to 995 existing SGAs ($SGA=1$). When assuming a uniform prior (i.e. each of these SGAs in this tumor influencing the MPL differential expression equally), the prior probability for each SGA can be calculated as, $1.0/995=0.001$. Considering this low prior probability, reaching a posterior probability of 0.01 after seeing the data is still a considerable positive update.

In addition, 14% ($n=28$) of the 201 eQTL PS-drivers, and 31% ($n=28$) out of the 90 IBI-DT PS-drivers were recognized as cancer drivers in CGC. Moreover, out of 90 IBI-DT PS-drivers, 16 were low-mutation or rare SGAs ($MF < 0.01$), in contrast to only one low-mutation SGA among 201 eQTL PS-drivers (Fig. 4C). Both methods identified AKT1 as a rare SGA. Among the 16 rare SGAs identified by IBI-DT, three are well-known cancer drivers, highlighting IBI-DT's capability to detect rare but significant SGAs.

Discovering conjunction and disjunction pairs through IBI-DT

Oncogenesis is characterized by a complex interplay of genetic alterations in key signaling pathways [33]. This complexity is further illustrated by the observed phenomenon of mutual exclusivity, where genetic alterations impacting members of the same pathway rarely co-occur in an individual tumor [14, 34, 35]. Meanwhile, numerous studies have pointed to co-mutation as a key factor influencing the progression of cancers driven by oncogenes [36]. A better understanding of the underlying signaling network will help develop targeted and precise therapies for treating

Table 1. Top 50 SGAs identified via the IBI-DT method.

Genes	Mutation frequency	Odds ratio	P-value	Rank of P-value	Number of appearances	Rank of number of appearances	Known or novel
PIK3CA	0.207	1.81	1.32E-11	20	184	1	Known
PTEN	0.114	1.75	3.48E-07	173	120	2	Known
APC	0.09	0.47	1.72E-06	219	116	3	Known
TP53	0.438	0.65	3.06E-08	125	115	4	Known
AKT1	0.009	3.22	2.41E-04	645	91	5	Known
PIK3R1	0.036	1.93	2.41E-04	649	76	6	Known
LRP1B	0.15	0.66	3.09E-04	695	67	7	Known
GATA3	0.038	2.41	1.82E-07	157	57	8	Known
PHB	0.016	5.87	2.02E-14	7	52	9	Novel
TTC7B	0.009	2.25	1.52E-02	2006	47	10	Novel
CSMD1	0.141	0.69	1.58E-03	1149	47	10	Novel
SYNE1	0.101	0.64	1.31E-03	1084	45	12	Novel
MAP3K1	0.035	1.66	7.01E-03	1676	44	13	Known
TTN	0.289	0.96	6.47E-01	11,922	42	14	Novel
RUNX1	0.009	2.03	4.23E-02	2766	41	15	Known
CDKN2A	0.166	0.72	2.27E-03	1265	40	17	Known
ZNF652	0.018	4.37	4.52E-11	26	40	17	Novel
USP13	0.066	0.47	3.30E-05	349	40	17	Novel
CHIC2	0.012	1.92	3.27E-02	2502	35	19	Known
SLC43A2	0.004	4.11	4.71E-03	1516	34	20	Novel
ATAD1	0.03	0.54	2.03E-02	2194	34	20	Novel
PNMT	0.046	2.92	1.46E-12	15	33	22	Novel
HUWE1	0.045	0.79	2.73E-01	6923	33	22	Novel
SRSF11	0.006	3.15	4.27E-03	1468	30	24	Novel
YIPF6	0.003	3.2	4.77E-02	2880	29	26	Novel
ARID1A	0.076	1.55	9.55E-04	995	29	26	Known
CCPG1	0.005	2.72	2.76E-02	2400	29	26	Novel
DHX40	0.023	4.79	2.45E-15	4	29	26	Novel
NTHL1	0.014	3.84	1.78E-07	156	28	30	Known
CSMD3	0.18	1.01	9.61E-01	15,437	28	30	Known
POF1B	0.006	3.15	4.27E-03	1468	27	32	Novel
IL33	0.004	2.56	1.00E-01	3934	27	32	Novel
SLC24A1	0.008	1.05	8.34E-01	14,472	26	34	Novel
AADACL4	0.005	1.07	7.82E-01	13,770	26	34	Novel
PTRH2	0.02	5.48	6.08E-16	2	25	35	Novel
RBM14	0.026	1.98	1.11E-03	1028	24	38	Novel
FAM9A	0.005	2.57	4.55E-02	2819	24	38	Novel
RIDA	0.05	1.91	2.56E-05	322	24	38	Novel
CTNNB1	0.039	1.09	6.26E-01	11,684	24	38	Known
IDH1	0.01	1.76	8.63E-02	3646	24	38	Known
PPP1R1B	0.043	2.9	6.28E-12	18	24	38	Novel
SLC16A8	0.004	3.86	3.69E-03	1430	23	44	Novel
EIF4EBP1	0.047	1.94	3.20E-05	345	23	44	Novel
MAP3K9	0.014	1.59	1.07E-01	4078	23	44	Novel
SDK2	0.033	2.26	6.82E-06	252	23	44	Novel
PGAP3	0.048	2.92	5.17E-13	13	22	48	Novel
PIWIL1	0.017	1.67	5.45E-02	3052	22	48	Novel
GPRASP1	0.016	1.15	6.51E-01	11,970	22	48	Novel
TBC1D3P1-DHX40P1	0.023	4.96	4.98E-16	1	22	48	Novel
CD1C	0.027	1.96	1.00E-03	1004	22	48	Novel

Note: If the gene is included in the COSMIC database, it is classified as "Known"; otherwise, it is classified as "Novel".

different types of cancer. However, it has been challenging to accurately reveal the functional interactions among cancer drivers. We applied IBI-DT to derive driver partners or pairs that potentially work together or complement each other in regulating downstream traits of interest, such as the DEG MPL. More specifically, IBI-DT derives the β statistic that tabulates how often SGA pairs appear together in the bootstrapped-derived trees. To understand the significance of β for each pair of SGAs, we performed a random node experiment where the SGAs were selected randomly as nodes in the decision tree. In this experiment, the maximum

β value of 2 indicates that pairs with $\beta > 2$ are unlikely to occur by random chance (Fig. 5A). While the total number of pairs analyzed in the IBI-DT was 160,289, among them 3720 pairs exceeded a β value of 2, as illustrated in Fig. 5A. The β value of the top pair, PIK3CA and APC, was 109. We focused on the top 50 pairs whose β is 19 or higher across 200 IBI-DT trees, which is about 10 times of the value of the highest β from random chance (Fig. 5A). As supported by the current literature, among the top 50 pairs, 16 were previously indicated for their potential interactions and 34 were novel findings (Table 2). The known pairs provide support for

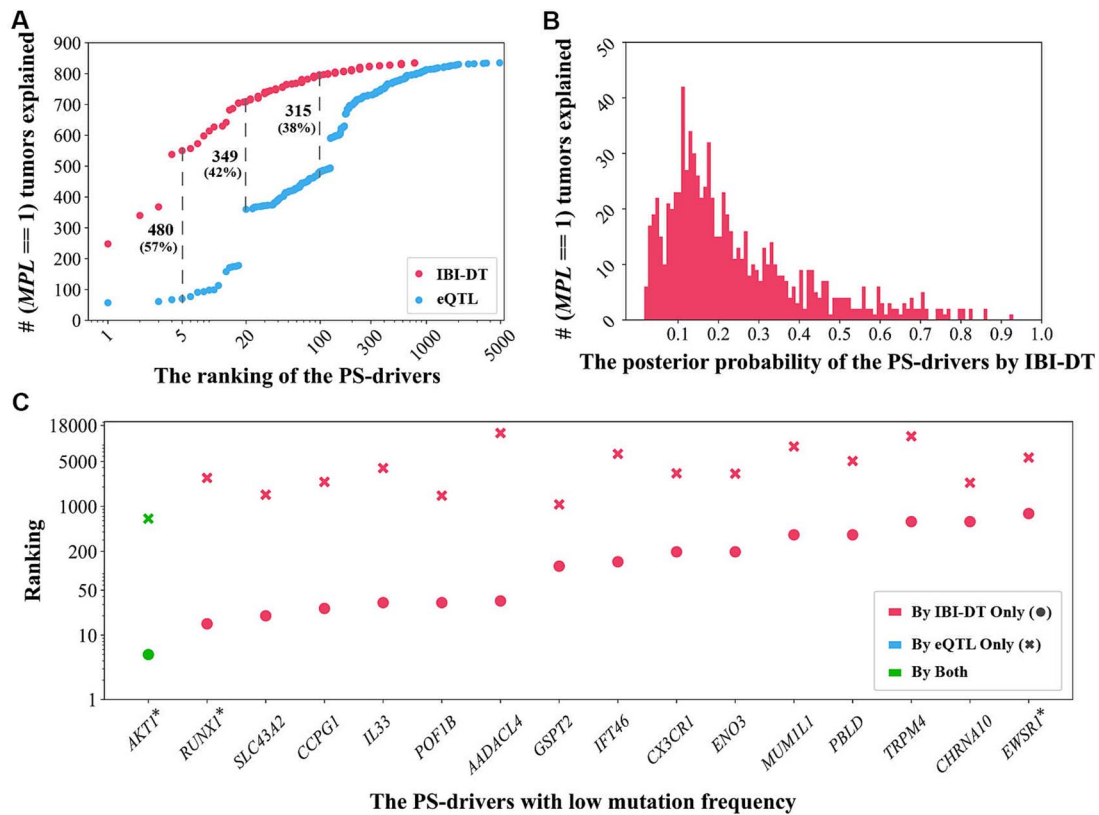


Figure 4. PS analysis for MPL = 1 tumors. (A) The number of MPL = 1 tumors explained by 90 IBI-DT PS-drivers indicated by red dots and 201 eQTL PS-drivers illustrated by blue dots, respectively. By rank 5, IBI-DT covers 550 samples, while eQTL covers 70 samples, resulting in a difference of 480 samples, or 57% of the MPL = 1 samples. Similarly, the sample coverage difference at ranks 20 and 100 is 42% and 38%, respectively. (B) A histogram of the individual-level posterior probabilities of the PS-drivers assigned by IBI-DT to each of the 835 tumors with MPL differential expression. (C) The ranking of rare PS-drivers (with a MF of 0.01 or less) identified by either IBI-DT, eQTL, or both. “x” indicates the ranking by IBI-DT while “#” represents the ranking by eQTL. “*” represents well-known CGC cancer drivers. Red colors show the low-mutation SGAs ($n = 15$) identified from IBI-DT PS-drivers only; blue colors indicate the low-mutation SGAs identified from eQTL PS-drivers only ($n = 0$); green colors suggest the low-mutation SGAs identified from both IBI-DT and eQTL PS-drivers ($n = 1$).

IBI-DT being able to identify known driver interactions (Table 2). The novel pairs suggest the potential of IBI-DT to help inform our understanding of driver interactions in the cancer signaling network (Table 2).

In Fig. 5B, a red edge represents a conjunctive interaction between SGA nodes, while a green edge indicates a disjunctive interaction. Figure 5B shows that out of 50 branch pairs, 11 pairs are connected conjunctively. Notably, among these conjunctively linked pairs, four specific interactions—PIK3CA and APC; PIK3CA and USP13; PTEN and ATAD1; and PTEN and PIK3R1—have been reported by literature, especially in the context of cancer initiation and development. First, it was found that 86% of colon tumors with mutations in the PIK3CA gene also contain mutations in the APC gene [37]. Interestingly, the combination of active PI3K and APC allelic loss leads to increased tumor multiplicity, bigger tumor sizes, and a more severe phenotype characterized by reduced differentiation [37]. Secondly, in 29% of high-grade serous ovarian cancers, the USP13 gene shows co-amplification with PIK3CA, and its increased expression is highly associated with a poor prognosis [38]. Thirdly, loss of ATAD1, a gene adjacent to and frequently co-deleted with PTEN, induces cancer cell susceptibility to apoptosis due to proteasome dysfunction and correlates with improved survival in cancer patients [39]. Fourthly, a notable co-occurrence of heterozygous PTEN mutations alongside mutations in either PIK3R1 or PIK3R2, was identified in endometrial cancers [40].

In Fig. 5B, out of 50 branch pairs, 39 are disjunction pairs. Nine out of these 39 disjunction pairs have been suggested by previous studies, while others are novel findings and have not been previously reported. Among these previously reported interactions, PTEN and TP53 mutations were not only reported to be mutually exclusive in the same tumor sample, but also their functional crosstalk has been suggested [41, 42]. Oliveira et al. conducted a study showing a total of 25 mutually exclusive events among three pairs of genes, where the PIK3CA-SYNE1 gene pair had 10 such events in ovarian clear cell carcinoma [43]. Other pairs, including PIK3CA and PIK3R1; PIK3CA and PTEN; PIK3CA and AKT1; PIK3CA and GATA3; PIK3CA and TP53; and PIK3CA and LRP1B, have also been reported to show mutual exclusivity in prior studies [44–50]. These findings highlight the significance of mutual exclusivity patterns in understanding cancer pathways and their regulatory mechanisms [44–50].

Discussion

We introduced a novel, robust and generalized method, IBI-DT, for discovering genomic variants/genes and their interactions that bear potentially functional impact on different phenotypes, e.g. DEGs, disease status, corresponding to different settings (e.g. eQTL analysis, GWAS analysis). By combining IBI with decision tree learning and bootstrapping [28], IBI-DT offers the following advantages: (i) subgroup-aware discovery vs. population-level

Table 2. Top 50 branch pairs identified via the IBI-DT method.

Parent gene	Child gene	Interaction	Number of appearances as partners	Known or novel
PIK3CA	APC	Conj.	109	Known [37]
PIK3CA	TP53	Disj.	68	Known [50]
PIK3CA	AKT1	Disj.	68	Known [48]
PIK3CA	PTEN	Disj.	57	Known [47]
PIK3CA	PIK3R1	Disj.	54	Known [46]
PTEN	TP53	Disj.	42	Known [50]
PIK3CA	USP13	Conj.	39	Known [38]
PIK3CA	TTC7B	Conj.	39	Novel
PIK3CA	RUNX1	Disj.	38	Novel
PHB	PIK3CA	Disj.	38	Novel
PIK3CA	LRP1B	Disj.	37	Known [50]
PTEN	PIK3CA	Disj.	37	Known [45]
PIK3CA	SYNE1	Disj.	36	Known [43]
PTEN	ATAD1	Conj.	34	Known [39]
PTEN	AKT1	Disj.	34	Known [45]
PTEN	CHIC2	Disj.	31	Novel
PIK3CA	GATA3	Disj.	30	Known [49]
PIK3CA	SLC43A2	Disj.	29	Novel
USP13	APC	Disj.	26	Novel
PTEN	PIK3R1	Conj.	25	Known [40]
PHB	APC	Disj.	25	Novel
PIK3CA	YIPF6	Disj.	25	Novel
PIK3CA	PPP1R1B	Disj.	24	Novel
PIK3CA	RIDA	Conj.	24	Novel
PIK3CA	MAP3K1	Disj.	24	Novel
PIK3CA	IL33	Disj.	24	Novel
PHB	PTEN	Disj.	23	Novel
AKT1	TP53	Disj.	23	Novel
PIK3CA	SLC24A1	Conj.	23	Novel
PIK3CA	HUWE1	Conj.	23	Novel
PIK3CA	POF1B	Disj.	22	Novel
PTEN	LRP1B	Disj.	22	Novel
PIK3CA	AADACL4	Disj.	21	Novel
PTRH2	PIK3CA	Disj.	21	Novel
PIK3CA	CCPG1	Disj.	21	Novel
PIK3CA	GPRASP1	Conj.	21	Novel
PIK3CA	TTN	Disj.	21	Novel
PIK3CA	PNMT	Disj.	21	Novel
PIK3CA	PIGA	Disj.	21	Novel
DHX40	PIK3CA	Disj.	20	Novel
AKT1	LRP1B	Disj.	20	Novel
PIK3CA	SH3GL2	Conj.	20	Novel
PHB	TP53	Disj.	20	Novel
APC	TTC7B	Disj.	20	Novel
PIK3CA	EXPH5	Conj.	20	Novel
PIK3CA	PNPLA2	Disj.	19	Novel
PIK3CA	ATP2A3	Disj.	19	Novel
PTEN	APC	Disj.	19	Novel
PIK3CA	SLC16A8	Disj.	19	Novel
AKT1	PTEN	Disj.	19	Known [45]

Note: "Conj." Indicates "Conjunctive" while "Disj." represents "Disjunctive".

inference: Unlike traditional population-based approaches (e.g. eQTL or MutSigCV) [5], IBI-DT explicitly models inter-patient heterogeneity by using bootstrapped decision trees to define genetically coherent or patient-like-me subgroups. This allows us to detect low-frequency variants (e.g. SGA) that are relevant within specific subgroups but would be missed by whole-cohort models. The use of bootstrap aggregation further improves robustness by sampling diverse subsets of the data, reducing sensitivity to outliers and cohort imbalance, and stabilizing the discovery of subgroup-specific SGAs across replicates. (ii) Bayesian inference for discovery, guided by marginal likelihood,

rather than prediction using heuristics: standard decision trees and random forests are built for prediction and use heuristic split criteria such as Gini impurity or information gain [17–20]. In contrast, IBI-DT is designed for discovery. It selects SGAs at each node using the Bayesian marginal likelihood, which quantifies how well a candidate SGA explains DEG variation within a patient subgroup, providing a principled measure of regulatory relevance. To prevent overfitting, especially at deeper nodes, IBI-DT applies a depth-based penalty by placing a prior over the tree structure, discouraging overly complex partitions. This regularization differs from the *ad hoc* pruning used in standard trees and from Bayesian

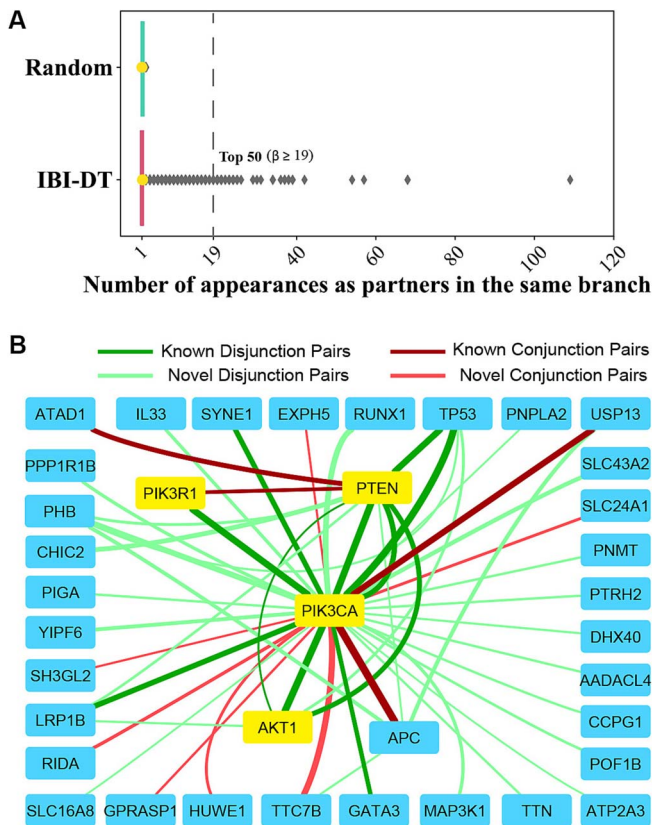


Figure 5. The potentially functional interactions identified by IBI-DT among cancer drivers (DEG MPL). (A) Comparison of the β between Random and IBI-DT methods. (B) IBI-DT network for the top 50 branch pairs including both conjunction and disjunction pairs. Yellow highlights the well-known drivers of DEG MPL.

models like Bayesian Additive Regression Trees [51], which place priors over entire trees and sample from posterior distributions for outcome prediction. IBI-DT, by contrast, applies Bayesian principles locally and deterministically to guide subgrouping and posterior estimation. Its goal is not to predict phenotypes or quantify uncertainty, but to identify interpretable and potentially functional regulatory patterns across heterogeneous populations. (iii) Personalized inference using individualized posterior probabilities: after aggregating subgroup-level statistics across bootstrapped trees, IBI-DT derives individualized posterior probabilities for each SGA per patient. These personalized scores reflect each tumor's unique mutational background and provide PS driver prioritization, a feature not offered by standard tree models or classical driver detection methods. (iv) Mechanistic interpretability and interaction discovery: the tree-based structure naturally supports interpretation of how driver mutations shape downstream expression changes, capturing logical interactions (conjunctive and disjunctive) inferred from tree paths that regulate gene expression patterns. These expression-informed interacting relationships are lacking in classic driver detection methods or statistical co-mutation methods such as DISCOVER or MEMO. [13, 14]. As a generalized framework, it can be applied to any discrete trait of interest including both clinical traits (i.e. disease status) and molecular traits (i.e. gene expression pattern, metabolomics); the upstream factors could be genomic variants, gene-level mutations, epigenomic factors, other factors of interest, or some combination of these factors.

We applied IBI-DT in the cancer domain to estimate cancer drivers from SGAs that regulate downstream gene expression (e.g. DEGs). Considering the discrete nature of SGA and DEG in our data, we used Fisher's exact test for eQTL analysis of the two selected DEGs; as a population-based approach that models two discrete variables, Fisher's exact test [23] served as an excellent control method to compare to the IBI-DT approach. Comparing to its population-based counterpart approach of Fisher's-exact-test-based eQTL [23], IBI-DT showed better performance in several tasks. For two different traits of target DEGs and across 10 different datasets ranging from ~500 to ~5000 samples, it identified known cancer drivers consistently better than eQTL. IBI-DT was able to consistently identify many key drivers with much higher α than the background/passenger SM. It was able to rank four well-known key drivers of a common cancer marker gene, MPL, at the very top (#1, 2, 5, 6) among the 17K mutations. In addition, it identified many well-known and novel cancer drivers with high α , when compared to the CGS database. This makes sense when considering that MPL is a common marker for many cancers. Moreover, IBI-DT was able to rank low-frequency mutations better than eQTL, probably due to its focus in patients-like subpopulations. In addition, the potential drivers identified by IBI-DT via individualized inference better explained the differential expression status of MPL in individual tumors. Moreover, IBI-DT was able to identify cancer drivers that function in disjunctive or conjunctive ways to regulate downstream gene expression patterns. Such information may lead to a better understanding of signaling pathways that underlie cancer development. These differences highlight the methodological distinction: eQTL emphasizes marginal associations between SGAs and gene expression across the population, whereas IBI-DT prioritizes SGAs that act in combinatorial or context-specific regulatory patterns within subgroups and at the level of an individual. As a result, IBI-DT may detect interacting or low-frequency drivers that would be overlooked by population-level marginal effect tests.

Limited tools are currently available for identifying the functional interactions among cancer drivers. Yet, how cancer drivers coordinate with each other is key to understanding the dysfunction in the underlying signal network leading to cancer formation and progression. IBI-DT was able to identify cancer driver that function disjunctively or conjunctively. In this paper, we focused on identifying and understanding the cancer driver pairs but interactions among three or more cancer drivers can be readily extracted from decision trees generated by IBI-DT. Some of the driver pairs are well-known supported by the literature, while others are novel and may be worthy of further investigation. Also importantly, IBI-DT is efficient in computing the high-order interactions embedded in a tree structure, thus allowing its usage with relatively large genomic datasets.

Limitations of IBI-DT include that it does not currently consider or adjust for the potential confounders of nongenomic factors such as age and sex, and it does not model continuous traits. When IBI-DT is applied to reveal the statistical relationships between germline variants and more complex diseases such as hypertension, the potential influence of confounding is likely much stronger than that of SM and cancer. Future extensions of IBI-DT aim to address non-genomic factors and model continuous traits. We believe the generalized framework of IBI-DT and its extensions have the potential to reveal novel insights to genomic factors and their interactions influencing the development and progression of many diseases, other than cancer.

Key Points

- By focusing on each individual and their “patients-like-me” subgroup, IBI-DT outperforms population-level methods in identifying key genomic factors for the trait of interest.
- IBI-DT efficiently detects high-order genetic interactions with potential functional impact by leveraging GPU acceleration and optimized matrix computations.
- In cancer applications, IBI-DT outperforms population-based approaches, identifying both known and novel cancer drivers (including those with low mutation frequency) and their interactions that potentially regulate downstream gene expression patterns.
- As a general framework, IBI-DT is applicable to diverse discrete trait (clinical or molecular), with upstream factors ranging from genomic variants and gene-level mutations to epigenomic factors or combinations thereof.

Supplementary data

Supplementary data are available at Briefings in Bioinformatics online.

Conflict of interest: None declared.

Funding

This work was supported by the National Heart, Lung and Blood Institute [K01HL161538 and R03HL168984 to JL] and [R01HL164835 to GFC], and the National Library of Medicine [R01LM012011 to XL].

Data availability

The TCGA data is publicly available on <https://portal.gdc.cancer.gov/>. The source codes, and the results of MPL are available at <https://github.com/asadcf/IBI-DT>.

References

- Weir B, Zhao X, Meyerson M. Somatic alterations in the human cancer genome. *Cancer Cell* 2004;**6**:433–8. <https://doi.org/10.1016/j.ccr.2004.11.004>.
- Chin L, Hahn WC, Getz G. et al. Making sense of cancer genomic data. *Genes Dev* 2011;**25**:534–55. <https://doi.org/10.1101/gad.2017311>.
- Sondka Z, Bamford S, Cole CG. et al. The COSMIC cancer gene census: Describing genetic dysfunction across all human cancers. *Nat Rev Cancer* 2018;**18**:696–705. <https://doi.org/10.1038/s41568-018-0060-1>.
- Youn A, Simon R. Identifying cancer driver genes in tumor genome sequencing studies. *Bioinformatics* 2011;**27**:175–81. <https://doi.org/10.1093/bioinformatics/btq630>.
- Lawrence MS, Stojanov P, Polak P. et al. Mutational heterogeneity in cancer and the search for new cancer-associated genes. *Nature* 2013;**499**:214–18.
- Bashashati A, Haffari G, Ding J. et al. DriverNet: uncovering the impact of somatic driver mutations on transcriptional networks in cancer. *Genome Biol* 2012;**13**:R124. <https://doi.org/10.1186/gb-2012-13-12-r124>.
- Leiserson MDM, Vandin F, Wu HT. et al. Pan-cancer network analysis identifies combinations of rare somatic mutations across pathways and protein complexes. *Nat Genet* 2015;**47**:106–14. <https://doi.org/10.1038/ng.3168>.
- Li Y, Zhang SW, Xie MY. et al. PhenoDriver: interpretable framework for studying personalized phenotype-associated driver genes in breast cancer. *Brief Bioinform* 2023;**24**:bbad291. <https://doi.org/10.1093/bib/bbad291>.
- Zhang T, Zhang SW, Xie MY. et al. A novel heterophilic graph diffusion convolutional network for identifying cancer driver genes. *Brief Bioinform* 2023;**24**:bbad137. <https://doi.org/10.1093/bib/bbad137>.
- Li H, Han Z, Sun Y. et al. CGMega: explainable graph neural network framework with attention mechanisms for cancer gene module dissection. *Nat Commun* 2024;**15**:5997. <https://doi.org/10.1038/s41467-024-50426-6>.
- Cai C, Cooper GF, Lu KN. et al. Systematic discovery of the functional impact of somatic genome alterations in individual tumors through tumor-specific causal inference. *PLoS Comput Biol* 2019;**15**:e1007088. <https://doi.org/10.1371/journal.pcbi.1007088>.
- Rahman MA, Cai C, Bo N. et al. An individualized Bayesian method for estimating genomic variants of hypertension. *BMC Genomics* 2023;**23**:863. <https://doi.org/10.1186/s12864-023-09757-9>.
- Canisius S, Martens JWM, Wessels LFA. A novel independence test for somatic alterations in cancer shows that biology drives mutual exclusivity but chance explains most co-occurrence. *Genome Biol* 2016;**17**:261. <https://doi.org/10.1186/s13059-016-1114-x>.
- Ciriello G, Cerami E, Sander C. et al. Mutual exclusivity analysis identifies oncogenic network modules. *Genome Res* 2012;**22**:398–406. <https://doi.org/10.1101/gr.125567.111>.
- Buntine W. Learning classification trees. *Stat Comput* 1992;**2**:63–73. <https://doi.org/10.1007/BF01889584>.
- Breiman L, Friedman JH, Olshen RA. et al. *Classification and regression trees*. 1st ed. New York, NY: Chapman and Hall/CRC, 1984. Reprinted 2017. <https://doi.org/10.1201/9781315139470>.
- Breiman L. Random forests. *Mach Learn* 2001;**45**:5–32. <https://doi.org/10.1023/A:1010933404324>.
- Kotsiantis SB. Decision trees: A recent overview. *Artif Intell Rev* 2013;**39**:261–83. <https://doi.org/10.1007/s10462-011-9272-4>.
- Raileanu LE, Stoffel K. Theoretical comparison between the Gini index and information gain criteria. *Ann math. Artif Intell* 2004;**41**:77–93.
- Suthaharan S, Suthaharan S. Decision tree learning. *Machine learning models and algorithms for big data classification: thinking with examples for effective learning* 2016;**36**:237–69. https://doi.org/10.1007/978-1-4899-7641-3_10.
- Yang HC, Lin CW, Chen CW. et al. Applying genome-wide gene-based expression quantitative trait locus mapping to study population ancestry and pharmacogenetics. *BMC Genomics* 2014;**15**:319. <https://doi.org/10.1186/1471-2164-15-319>.
- Wang X, Cheng A, Consortium P. et al. An eQTL and mQTL study of 147 genetic variants associated with prostate cancer susceptibility. *Cancer Res* 2018;**78**:1218.
- Fisher RA. Statistical Methods for Research Workers. In: Kotz S, Johnson NL (eds.), *Breakthroughs in Statistics: Methodology and Distribution*. New York, NY: Springer, 1992, p. 66–70.
- Pulikkan JA, Madera D, Xue L. et al. Thrombopoietin/MPL participates in initiating and maintaining RUNX1-ETO acute myeloid leukemia via PI3K/AKT signaling. *Blood* 2012;**120**:868–79. <https://doi.org/10.1182/blood-2012-03-414649>.

25. Eferl R, Wagner EF. AP-1: A double-edged sword in tumorigenesis. *Nat Rev Cancer* 2003;**3**:859–68. <https://doi.org/10.1038/nrc1209>.
26. Wang Z, Townley SL, Zhang S. et al. FOXA2 rewires AP-1 for transcriptional reprogramming and lineage plasticity in prostate cancer. *Nat Commun* 2024;**15**:4914. <https://doi.org/10.1038/s41467-024-49234-9>.
27. Heckerman D, Meek C, Cooper GF. A Bayesian approach to causal discovery. *Innovations in Machine Learning: Theory and Applications*. Berlin, Heidelberg: Springer, 2006, p. 1–28.
28. Efron B. Bootstrap Methods: Another Look at the Jackknife. In: Kotz S, Johnson NL (eds.), *Breakthroughs in Statistics: Methodology and Distribution*. New York, NY: Springer, 1992, p. 569–93.
29. Hoeting JA, Madigan D, Raftery AE. et al. Bayesian model averaging: A tutorial with comments by M. Clyde, David Draper and E. I. George, and a rejoinder by the authors. *Stat Sci* 1999;**14**:382–417.
30. Gelman A, Carlin JB, Stern HS. et al. *Bayesian Data Analysis*. London: Chapman and Hall/CRC, 1995.
31. Vogelstein B, Papadopoulos N, Velculescu VE. et al. Cancer genome landscapes. *Science* 1979;**339**:1546–58. <https://doi.org/10.1126/science.1235122>.
32. Martínez-Jiménez F, Muñíos F, Sentís I. et al. A compendium of mutational cancer driver genes. *Nat Rev Cancer* 2020;**20**:555–72. <https://doi.org/10.1038/s41568-020-0290-x>.
33. Sanchez-Vega F, Mina M, Armenia J. et al. Oncogenic Signaling pathways in the cancer genome atlas. *Cell* 2018;**173**:321–337.e10. <https://doi.org/10.1016/j.cell.2018.03.035>.
34. Vandin F, Upfal E, Raphael BJ. De novo discovery of mutated driver pathways in cancer. *Genome Res* 2012;**22**:375–85. <https://doi.org/10.1101/gr.120477.111>.
35. Lu S, Lu KN, Cheng SY. et al. Identifying driver genomic alterations in cancers by searching minimum-weight, mutually exclusive sets. *PLoS Comput Biol* 2015;**11**:e1004257. <https://doi.org/10.1371/journal.pcbi.1004257>.
36. Jiang L, Yu H, Ness S. et al. Comprehensive analysis of Co-mutations identifies cooperating mechanisms of tumorigenesis. *Cancers (Basel)* 2022;**14**:415.
37. Deming DA, Leystra AA, Nettekoven L. et al. PIK3CA and APC mutations are synergistic in the development of intestinal cancers. *Oncogene* 2014;**33**:2245–54. <https://doi.org/10.1038/onc.2013.167>.
38. Han C, Yang L, Choi HH. et al. Amplification of USP13 drives ovarian cancer metabolism. *Nat Commun* 2016;**7**:13525. <https://doi.org/10.1038/ncomms13525>.
39. Winter JM, Fresenius HL, Cunningham CN. et al. Collateral deletion of the mitochondrial AAA+ ATPase ATAD1 sensitizes cancer cells to proteasome dysfunction. *elife* 2022;**11**:11. <https://doi.org/10.7554/eLife.82860>.
40. Cheung LWT, Hennessy BT, Li J. et al. High frequency of PIK3R1 and PIK3R2 mutations in endometrial cancer elucidates a novel mechanism for regulation of PTEN protein stability. *Cancer Discov* 2011;**1**:170–85. <https://doi.org/10.1158/2159-8290.CD-11-0039>.
41. Trotman LC, Pandolfi PP. PTEN and p53: who will get the upper hand? *Cancer Cell* 2003;**3**:97–9. [https://doi.org/10.1016/s1535-6108\(03\)00022-9](https://doi.org/10.1016/s1535-6108(03)00022-9).
42. Kurose K, Gilley K, Matsumoto S. et al. Frequent somatic mutations in PTEN and TP53 are mutually exclusive in the stroma of breast carcinomas. *Nat Genet* 2002;**32**:355–357. <https://doi.org/10.1038/ng1013>.
43. Oliveira DVNP, Schnack TH, Poulsen TS. et al. Genomic subclassification of ovarian clear cell carcinoma revealed by distinct mutational signatures. *Cancers (Basel)* 2021;**13**. <https://doi.org/10.3390/cancers13205242>.
44. Wang Y, Hou P, Yu H. et al. High prevalence and mutual exclusivity of genetic alterations in the phosphatidylinositol-3-kinase/Akt pathway in thyroid tumors. *J Clin Endocrinol Metab* 2007;**92**:2387–90. <https://doi.org/10.1210/jc.2006-2019>.
45. Gutierrez A, Sanda T, Grebliunaite R. et al. High frequency of PTEN, PI3K, and AKT abnormalities in T-cell acute lymphoblastic leukemia. *Blood* 2009;**114**:647–50. <https://doi.org/10.1182/blood-2009-02-206722>.
46. Chen L, Yang L, Yao L. et al. Characterization of PIK3CA and PIK3R1 somatic mutations in Chinese breast cancer patients. *Nat Commun* 2018;**9**:1357.
47. Saal LH, Holm K, Maurer M. et al. PIK3CA mutations correlate with hormone receptors, node metastasis, and ERBB2, and are mutually exclusive with PTEN loss in human breast carcinoma. *Cancer Res* 2005;**65**:2554–2559. <https://doi.org/10.1158/0008-5472.CAN-04-3913>.
48. Samuels Y, Waldman T. Oncogenic mutations of PIK3CA in human cancers. *Curr Top Microbiol Immunol* 2010;**347**:21–41. https://doi.org/10.1007/82_2010_68.
49. Jiang YZ, Da YK, Zuo WJ. et al. GATA3 mutations define a unique subtype of luminal-like breast cancer with improved survival. *Cancer* 2014;**120**:1329–37. <https://doi.org/10.1002/cncr.28566>.
50. Zhang Y, Yang W, Liu JS. et al. Characterizing mutually exclusive driver mutations in pan-cancer. *Proceedings - 2019 IEEE international conference on bioinformatics and biomedicine (BIBM)*. 2019, pp. 2013–2020. <https://doi.org/10.1109/BIBM47256.2019.8983324>.
51. Chipman HA, George EI, McCulloch RE. BART: Bayesian additive regression trees. *Ann Appl Stat* 2012;**6**:21–41.